SESSION 11

Valuation & Forecasting 1

SCALER SCHOOL OF TECHNOLOGY



Learning Outcomes

Session 11: Valuation & Forecasting – 1

By the end of the session, learners will be able to:

- **Explain** free cash flow (FCF), FCFF, FCFE, DCF, and valuation multiples.
- **Differentiate** FCFF from FCFE and P/E from Enterprise Value multiples.
- **Calculate** FCFF and FCFE from basic financial data.
- **Apply** DCF and multiples-based approaches to estimate company value.
- **Compare** FCFF and FCFE valuations and the impact of capital structure.
- **Analyze** how discount rates, growth assumptions, and multiples affect valuation.

Valuation & Forecasting

Valuation is the process of estimating **the economic value of an asset, business, or company.**

The value of an asset = The present value of the future economic benefits that investors expect to receive from it.

Valuation is based on:

✔Future cash flows
✔Earnings
✔Growth prospects
✔Risk
✔Capital requirements – debt or equity
✔Market valuations of comparable companies

Valuation & Forecasting

Approaches to Valuation

There are two broad ways of thinking about valuation.

❑**Intrinsic valuation –** What are the company's future cash flows worth today?
❑**Relative Valuation –** How much are similar companies being valued by the market?

Uses of Valuation:

- Investment decisions
- Mergers & Acquisitions (M&A)
- Initial Public Offer (IPO) pricing
- Sale or Purchase of Business
- Strategic decisions

It is important to note that **Market Price ≠ Intrinsic Value**

✔**Market price** is the price at which the asset is trading.
✔**Intrinsic value** is an estimate based on the inherent strength of an asset, i.e., its ability to generate future cash flows.

Valuation & Forecasting

Factors that affect the Value of a Business:

- **Cash flows:** How much cash can the business generate?
- **Growth:** How quickly can cash flows grow?
- **Risk:** How uncertain are future cash flows?
- **Investment requirements:** How much must the company reinvest?
- **Capital structure:** How much debt and equity does the company use?
- **Terminal value:** What is the value of the business beyond the explicit forecast period?

Free Cash Flow (FCF)

Accounting profit measures performance; free cash flow focuses on cash available after necessary investment.

Free Cash Flow (FCF) is the cash generated by a business after making the investments required to maintain and grow the business.

FCF = Operating Cash Flow − Capital Expenditure

Free Cash Flow (FCF)

ABC Manufacturing Ltd. reports the following for FY2026.

Particulars	₹ crore
Net Profit after Tax	**1,000**
Depreciation	200
Increase in Inventory	100
Increase in Trade Receivables	50
Increase in Trade Payables	40
Capital Expenditure – machinery	300

At first glance, it seems that the company earned ₹1,000 crore, so it has ₹1,000 crore available to distribute.

However, due to accrual accounting & other adjustments, that **Cash ≠ Profit**

Free Cash Flow (FCF)

The company reports a Net Profit of ₹1,000 crore.

Particulars	₹ crore
Net Profit after Tax	**1,000**
Depreciation	200
Increase in Inventory	100
Increase in Trade Receivables	50
Increase in Trade Payables	40
Capital Expenditure – machinery	300

But **net profit** is an **accounting measure,** not a **direct measure of cash generated.** It needs to be adjusted

Add back Depreciation of ₹200 crore, as no cash is paid for charging depreciation.

Therefore, we add **depreciation back to profit: ₹1,000 + ₹200 = ₹1,200 crore**

Free Cash Flow (FCF)

The company reports a Net Profit of ₹1,000 crore.

Particulars	₹ crore
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Capital Expenditure – machinery	300

Adjustment for Working Capital:

Working capital generally includes items such as:

- Inventory
- Trade receivables
- Trade payables

The key question is: **Did the change in working capital consume cash or generate cash?**

Increase in Inventory = cash outflow

Therefore, ₹100 crore to be subtracted: ₹1,200 - ₹100 = ₹1,100 crore

Free Cash Flow (FCF)

The company reports a Net Profit of ₹1,000 crore.

Particulars	₹ crore
Net Profit after Tax	**1,000**
Depreciation	200
Increase in Inventory	100
Increase in Trade Receivables	50
Increase in Trade Payables	40
Capital Expenditure – machinery	300

Adjustment for Trade Receivables:
Trade receivables increased by **₹50 crore. It signifies that** the company has made sales to customers but has **not yet collected the cash.**

Therefore, the increase in receivables represents **cash that is temporarily tied up with customers.**

Increase in Receivables = cash outflow

Therefore, ₹50 crore to be subtracted: ₹1,100 - ₹50 = ₹1,050 crore

Free Cash Flow (FCF)

The company reports a Net Profit of ₹1,000 crore.

Particulars	₹ crore
Net Profit after Tax	**1,000**
Depreciation	200
Increase in Inventory	100
Increase in Trade Receivables	50
Increase in Trade Payables	40
Capital Expenditure – machinery	300

Adjustment for Trade Payables:
Trade payables increased by **₹40 crore. It signifies that** the company has received goods/services **but has not yet paid the supplier.**

Therefore, the company has effectively **retained ₹40 crore of cash.**

the increase in receivables represents **cash that is temporarily tied up with customers.**

Increase in Payables = cash inflow

Therefore, ₹40 crore to be added: ₹ 1,050 = ₹40 = ₹1,090 crore

Free Cash Flow (FCF)

The company reports a Net Profit of ₹1,000 crore.

Particulars	₹ crore
Net Profit after Tax	**1,000**
Depreciation	200
Increase in Inventory	100
Increase in Trade Receivables	50
Increase in Trade Payables	40
Capital Expenditure – machinery	300

Calculating Operating Cash Flow:

Particulars	₹ crore	Cash-flow adjustment
Net Profit	1,000	Starting point
Add: Depreciation	+200	Non-cash expense
Less: Increase in Inventory	−100	Cash tied up
Less: Increase in Receivables	−50	Cash not yet collected
Add: Increase in Payables	+40	Cash retained
Operating Cash Flow	**1,090**	

Free Cash Flow (FCF)

The company reports a Net Profit of ₹1,000 crore.

Particulars	₹ crore
Net Profit after Tax	**1,000**
Depreciation	200
Increase in Inventory	100
Increase in Trade Receivables	50
Increase in Trade Payables	40
Capital Expenditure – machinery	300

Adjustment for Capital Investment:

Company purchased a new machinery for **₹300 crore.** This is **capital expenditure (CapEx).**

✔Unlike depreciation, **CapEx represents an actual cash payment.**

The company actually paid: **₹300 crore**

Capital Expenditure = cash inflow

Therefore, ₹300 crore to be subtracted = ₹ 1,090 - ₹300 = ₹790 crore

Free Cash Flow (FCF)

The company reports a Net Profit of ₹1,000 crore.

Particulars	₹ crore
Net Profit after Tax	**1,000**
Depreciation	200
Increase in Inventory	100
Increase in Trade Receivables	50
Increase in Trade Payables	40
Capital Expenditure – machinery	300

Free Cash Flow (FCF): Operating Cash Flow - Capital Expenditure

FCF = ₹ 1,090 - ₹300 = ₹790 crore

Free Cash Flow to Firm (FCFF)

Free Cash Flow to Firm (FCFF) is the cash generated by a company's operations that is available to **all providers of capital:**

✔Equity shareholders
✔Debt holders

It is the **cash flow generated by the business before payments to debt providers and equity shareholders.**

$$FCFF = EBIT(1 - T) + Depreciation - CapEx - \Delta NWC$$

Where:

- **EBIT** = Earnings Before Interest and Tax
- **T** = Corporate tax rate
- **Depreciation** = Non-cash expense
- **CapEx** = Capital expenditure
- **ΔNWC** = Change in operating net working capital

An alternative formulation using CFO is possible, but the EBIT-based formula is particularly useful for valuation because it removes the effect of financing decisions.

Free Cash Flow to Equity (FCFE)

Free Cash Flow to Equity (FCFE) measures the cash available specifically to **equity shareholders after meeting the cash-flow consequences of debt financing.**

A commonly used formula is:

$$\boxed{FCFE = NI + Depreciation - CapEx - \Delta NWC + NetBorrowing}$$

Where:

$$\boxed{Net\ Borrowing = New\ Debt\ Raised - Debt\ Repaid}$$

Alternatively:

$$\boxed{FCFE = FCFF - Interest(1 - T) + Net\ Borrowing}$$

This second formula is extremely useful for understanding the relationship between FCFF and FCFE.

Free Cash Flow to Equity (FCFE)

Example:

₹ crore	FY2020–21	FY2021–22	FY2022–23	FY2023–24	FY2024–25
Profit after tax	33,388	38,449	42,303	46,099	48,797
Effective tax rate	25.5%	25.6%	25.7%	25.8%	25.3%
EBIT margin	25.9%	25.3%	24.1%	24.6%	24.3%
Revenue	1,64,177	1,91,754	2,25,458	2,40,893	2,55,324
EBIT = Revenue × EBIT margin	42,520	48,514	54,336	59,260	62,045
D&A	4,929	5,337	5,586	4,985	5,242
Operating cash flow	39,929	45,067	44,000	44,990	49,139
CapEx	3,119	2,876	3,155	3,063	2,650
FCFF ≈ CFO − CapEx	36,810	42,191	40,845	41,927	46,489
Cash interest / finance cost*	~600	~600	~700	778	796
Net borrowing	~0	~0	~0	~0	~0
FCFE ≈ CFO − CapEx − Interest + Net Borrowing	36,210	41,591	40,145	41,149	45,693

Free Cash Flow to Equity (FCFE)

Example:

₹ crore	FY2020–21	FY2021–22	FY2022–23	FY2023–24	FY2024–25
Revenue	5,39,238	7,88,743	9,74,864	10,00,122	10,71,174
EBITDA / EBDIT	97,580	1,23,684	1,53,920	1,78,290	1,83,422
D&A	26,572	29,782	40,303	50,832	53,136
EBIT = EBITDA − D&A	71,008	93,902	1,13,617	1,27,458	1,30,286
Approx. effective tax rate	~25%	~25%	~25%	~25%	~23%
NOPAT = EBIT × (1−Tax)	53,256	70,427	85,213	95,594	1,00,320
D&A	26,572	29,782	40,303	50,832	53,136
Less: CapEx	~1,28,000	~1,09,000	1,41,809	1,31,769	1,39,967
Less: ΔNWC / operating working-capital investment	~15,000	~10,000	~15,000	~6,366	~23,724
FCFF	~(63,172)	~(18,791)	~(31,293)	~8,291	~(10,235)
CFO — reported	—	—	1,15,032	1,58,788	1,78,703
Cash interest paid	—	—	~21,650	37,173	41,024
Net borrowing	—	—	~38,075	9,262	19,247
FCFE — cash-flow approach	—	—	~(10,352)	~(833)	17,?

Thank You!